

TUE

Name

Mat

Find the General Solution of the following ODEs

1. $ty' - 2y = t^3 e^{-t}$

$$y' - \frac{2}{t}y = t^2 e^{-t}$$

$$t^{-2}y' - 2t^{-3}y = e^{-t}$$

$$(t^{-2}y)' = e^{-t}$$

$$t^{-2}y = -e^{-t} + C$$

1. $ty' - 4y = 3t^{-2}$

$$y(t) = -\frac{1}{2}t^{-2} + Ct^4$$

lin $y' + p y = f$

not-sep,

$$p(t) = -\frac{2}{t} \int -\frac{2}{t} dt = -2 \ln(t)$$

$$v(t) = e^{-2 \ln(t)} = e^{-2 \ln(t)}$$

$$v(t) = e^{\ln(t^{-2})} = t^{-2}$$

$$y = -t^2 e^{-t} + Ct^2$$

$$y' - \frac{4}{t}y = 3t^{-3}$$

$$v(t) = e^{-4 \ln(t)} = t^{-4}$$

$$(t^{-4}y)' = 3t^{-7}$$

$$t^{-4}y = \int 3t^{-7} dt + C$$

$$t^{-4}y = -\frac{1}{2}t^{-6} + C$$

$$y = -\frac{1}{2}t^{-2} + Ct^4$$

$$\partial M / \partial y = 4ty \text{ equal so exact!}$$

$$\partial N / \partial t = 4ty$$

$$m dt + n dy = 0$$

$$\square (1 + 2ty^2) dt + (2y + 2t^2y + 3y^2) dy = 0 \text{ aka } y' = -(1 + 2ty^2) / (2y + 2t^2y + 3y^2)$$

$$\frac{\partial S}{\partial t} = 1 + 2ty^2 \Rightarrow S = t + t^2y^2 + h(y) \Rightarrow \begin{cases} \frac{\partial S}{\partial t} = 0 + 2t^2y + h'(y) \\ \frac{\partial S}{\partial y} = 2y + 2t^2y + 3y^2 \end{cases}$$

$$\text{so } h'(y) = 2y + 3y^2 \Rightarrow h(y) = y^2 + y^3$$

Gen sol is ~~XXXXXXXXXXXX~~

$$t + t^2y^2 + y^2 + y^3 = C$$

$$2. y' = -\frac{2t+y}{1+3y^2}$$

$$(t + 3y^2) dy + (2t + y) dt = 0$$

$$t^2 + ty + y^3 = C$$

$$\frac{\partial}{\partial t} (t + 3y^2) = 1 = \frac{\partial}{\partial y} (2t + y)$$

Exact

$$S = \int t + 3y^2 dy + h(t) = ty + y^3 + h(t)$$

$$\Rightarrow \frac{\partial S}{\partial t} = y + 0 + h'(t) = 2t + y \Rightarrow \begin{cases} h'(t) = 2t \\ h(t) = t^2 \end{cases}$$

$$\text{or } S = \int 2t + y dt + g(y) = t^2 + ty + g(y)$$

$$\Rightarrow \frac{\partial S}{\partial y} = 0 + t + g'(y) = t + 3y^2 \Rightarrow \begin{cases} g'(y) = 3y^2 \\ g(y) = y^3 \end{cases}$$